

Adverse Event Following Change in Brachytherapy Plaque-Cleaning Chemicals: A Case Series

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Keywords

Plaque-cleaning chemicals · Uveal melanoma · Brachytherapy · Ruthenium-106 · Silver chloride formation on plaque device

Abstract

Introduction: The brachytherapy plaques used for uveal melanoma treatment are reused following sterilization processes. Here, we report patients with adverse reaction following a new cleaning procedure for brachytherapy plaques. **Case Presentations:** The brachytherapy plaque sewn on the outer eyewall would be removed following the delivery of calculated radiation dose. They are reused following standard calibration testing and sterilization processes. From September 2021 to January 2022, 6 patients treated with ruthenium-106 plaque brachytherapy presented with severe unilateral pain, eyelid swelling, and conjunctival injection with hemorrhage 1–2 days following plaque insertion. The surgeons observed a frosted appearance of the plaques in contrast to the normal shiny look during removal. The unusual severe postoperative reactions and the frosted appearance of plaques led to further investigations. A change in sterilization procedures with the use of LifeClean™ instead of PeraSafe™ was noticed. Ruthenium-106 brachytherapy plaques contain a uniformly distributed radioactive source covered by a thin silver shield. A

test was performed by using two plates of pure silver that underwent its respective cleaning procedure using either PeraSafe™ or LifeClean™. **Conclusion:** The test results demonstrated formation of silver chloride with LifeClean™, while no such formation was demonstrated with PeraSafe™. We did not observe any new cases of severe postoperative reaction or frosting of plaques after changing back to PeraSafe™. Chlorine salt deposits probably contributed to severe inflammatory reaction of ocular surface.

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Published by S. Karger AG, Basel

Introduction

Uveal melanoma arises from melanocytes and represents the most common primary intraocular tumor in adults [1, 2]. Plaque brachytherapy has paved the way for eye- and vision-preserving treatment. In plaque brachytherapy, the radioactive isotopes deliver a highly concentrated radiation dose to the tumor. The radioactive plaque is sutured on the scleral surface overlying the tumor, and the plaque is removed after a defined period of time depending on tumor size and calculated radiation dose needed for treatment.

Potential complications and adverse events of plaque brachytherapy may be encountered. The potential

complications of high doses of radiation include cataract, retinopathy, and optic neuropathy [3–8]. Other side effects of brachytherapy include neovascular glaucoma, radiation-induced dry eye, scleral necrosis, and strabismus [3, 5].

The radioresistant sclera receives the highest dose of radiation during brachytherapy, but it rarely develops complications such as scleritis, episcleritis, and scleral necrosis [9]. Scleral necrosis following brachytherapy occurs rarely due to avascularity and hypocellularity of the sclera, occurs in the first 6 months, and is mostly detected in an asymptomatic patient [10]. The arteries in the choroid and episclera supply nutrition to the sclera and may predispose to the development of vasculitis or more devastating necrosis with destruction of structural components of the sclera [10, 11]. Inflammation of the sclera is present with more pain in comparison with scleral necrosis due to cellular infiltration and stimulation of nerve endings. In scleral necrosis, the necrotic area may appear over time as an ectasia with a thin layer of conjunctiva or episclera [12]. Most commonly, the scleral necrosis occurs months to years following brachytherapy [13–15]. In this case series, we report an inflammatory reaction of the eyelids and conjunctiva following the use of a new cleaning agent (LifeClean™) in the sterilization process of the plaques.

Case Presentations

From September 2021 to January 2022, 6 patients out of 11 with uveal melanoma treated with ruthenium-106 brachytherapy developed swollen eyelids and conjunctival injection, hemorrhage, and chemosis 1–2 days following plaque insertion. During the same period, only 1 patient was treated with iodine-125 brachytherapy plaque and did not develop any unusual or severe postoperative reaction. All the patients in this case series were Norwegians and treated at the Department of Ophthalmology, Oslo University Hospital. They were treated with the radiation dose and time as pre-calculated by the radiation physicists. Here, we present 4 cases in detail (4 patients gave written informed consent for publication of this case series and the accompanying image, and the other 2 patients were not able to give consent due to demise). The details of tumor size, brachytherapy plaques, and radiation doses are given in Table 1.

The first patient (case 1), a 56-year-old female, underwent treatment with ruthenium-106 plaque brachytherapy for a clinically diagnosed choroidal melanoma ($4.7 \times 5.1 \times 1.4$ mm) in the right eye superotemporal to the optic disc. On routine first postoperative day examination

following plaque insertion, the examination revealed severe eyelid swelling, conjunctival injections, and pain. During removal of the plaque and resuturing of the superior oblique rectus muscle 2 days following plaque insertion, the surgeon noticed whitish discharge in close proximity to the lateral rectus muscle. The patient received intravenous cloxacillin treatment. On routine first postoperative day examination following plaque removal, the patient still had eyelid swelling, severe conjunctival injections, and pain on palpation, but no longer had discharge. The intraocular pressure (IOP) was 21 mm Hg. A remarkable improvement was noted on day two following plaque removal, and the patient was discharged the following day on oral antibiotics (dicloxacillin) and combined topical steroids and antibiotics (Maxitrol® containing dexamethasone 0.1%, neomycin 3.5 mg/g, and polymyxin B sulfate 6,000 IU/g). No bacterial growth was detected in the culture of the discharge. She had satisfying tumor control at the 2-year follow-up, and the tumor height could not be measured on ultrasound. Her best-corrected visual acuity (BCVA) had decreased from 1.0 to 0.63 secondary to cataract.

The second patient (case 2), a 51-year-old female with biopsy verified left-sided choroidal melanoma (spindle type, gene *BAP1* negative) measuring $14.0 \times 14.0 \times 2.4$ mm located 1.5 disc diameter away from fovea, was treated with ruthenium-106 plaque brachytherapy. She developed unilateral ocular pain, conjunctival hemorrhage, and chemosis on the second postoperative day. IOP was 28 mm Hg. Due to the clinical suspicion of localized infection, she was treated with intravenous cefuroxime in addition to topical Maxitrol®. Following removal of the plaque on the third postoperative day as planned, the patient noticed immediate improvement and she was discharged the same day on topical Maxitrol® eye drops. At 2-year follow-up, the BCVA had decreased from 1.0 to 0.5 and tumor height had decreased to 1.3 mm.

The third patient (case 3), a 69-year-old male with a choroidal melanoma superotemporally in the right eye, underwent surgical treatment with ruthenium-106 plaque and partial loosening of superior rectus muscle. Tumor measurement revealed a size of $6.3 \times 6.2 \times 2.5$ mm. On the second postoperative day, the patient developed unilateral ocular pain and whitish discharge. IOP was 17 mm Hg. The examination revealed severe eyelid swelling and conjunctival chemosis. Due to a clinical suspicion of cellulitis, the patient was treated with intravenous cefuroxime. At the removal of the plaque at the pre-calculated timing, the superior rectus muscle was repositioned. No bacterial or fungal growth was detected in the whitish discharge culture. He was discharged the following day on topical Maxitrol® eye

Table 1. The details of uveal melanoma size and brachytherapy (ruthenium-106) are given

Case	Tumor height, mm	Tumor base, mm	Brachytherapy plaque	Plaque diameter, mm	Treatment time, h	Given dose, Gy	Dose rate, Gy/h	Scleral surface dose, Gy	Scleral dose 1 mm inside, Gy
1	1.4	4.7 × 5.1	CCA-2509	15	50.8	121.6	2.39	311	247
2	2.4	14.0 × 14.0	CCB-2868	20	64.7	101.9	1.58	352	275
3	2.5	6.3 × 6.2	CCA-2509	15	72.0	99.0	1.38	378	300
4	1.6	6.1 × 3.9	COB-1322	20 (notched)	65.4	120.0	1.86	328	257

drops. At 14-month follow-up, the tumor control was satisfactory and no height could be detected on ultrasound. He demonstrated a slight decrease of pretreatment BCVA from 0.5 to 0.4 due to increasing nuclear cataract.

The fourth patient (case 4), a 74 year-old female with a right-sided choroidal melanoma measuring 6.1 × 3.9 × 1.6 mm located close to the optic disc, was treated with ruthenium-106 plaque. The patient developed unilateral eyelid swelling, conjunctival injection, and chemosis in the treated eye on the second postoperative day. The treatment included topical Maxitrol® eye drops, and the brachytherapy plaque was removed the following day as planned. At 19-month follow-up, her BCVA had improved from 0.5 to 0.8; however, she had undergone cataract surgery prior to the follow-up. The tumor height had decreased to 1.1 mm.

The removal of the brachytherapy plaques relieved the symptoms in all patients. The brachytherapy plaques (ruthenium-106) appeared frosted at the removal in all these patients as shown in Figure 1.

Testing of Cleaning Agents

Both PeraSafe™ and LifeClean™ are oxidative antimicrobial agents, but with different types of oxidation agents. PeraSafe™ is based on a mildly basic solution of hydrogen peroxide, delivered as a solid substance producing percarbonate and peracetic acid when dissolved in water, in addition to containing citric acid as a complexing and pH-regulating agent, resulting in a pH of ca. 8. LifeClean™, on the other hand, is based on an acidic solution (pH = 2) of chlorine dioxide in water as the oxidation agent. Chlorine dioxide is only slightly soluble in water, but upon reaction forms chlorite and chlorate. The oxidative power of chlorine dioxide under acidic conditions is notably stronger than the peroxide in PeraSafe™ under mildly basic conditions; however, both are sufficiently strong to oxidize silver to its oxide. Given

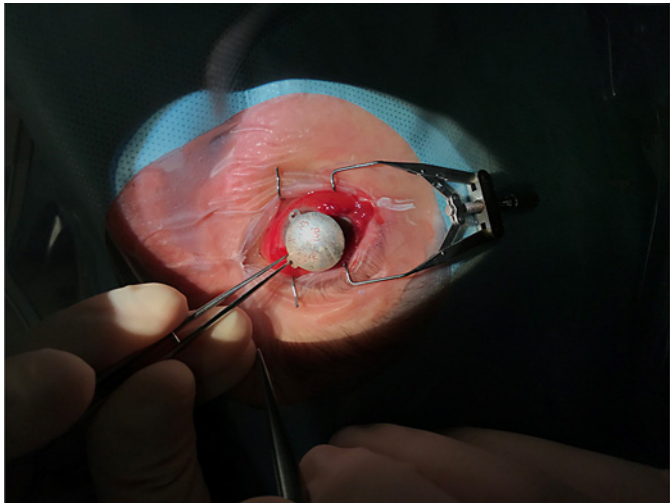


Fig. 1. This image was taken during surgical removal of brachytherapy plaque. It demonstrates the frosted appearance of the ruthenium-106 plaque in contrast to a normal, shiny appearance.

this, one would assume that both would be equally good as disinfectants for silver; however, they differ in their subsequent reactions. For PeraSafe™, one will encounter the catalytic power of silver to decompose the peroxides into water and oxygen, and hence effectively reduce its activity on silver. This reaction also aids in removing any excess peroxide before application to the eye. In addition, the mild pH conditions ensure that any silver oxide formed will act as a protective layer toward further oxidation. For LifeClean™, any such protective layer will dissolve under its acidic conditions, thus enabling further oxidation and roughening of the silver surface. Unlike for peroxides, silver does not catalytically decompose chlorite or chlorate, but rather forms slightly soluble compounds with silver (AgClO₂ and AgClO₃) in addition to insoluble silver chloride as a by-product. Any remaining active ingredients on the surface will likely be

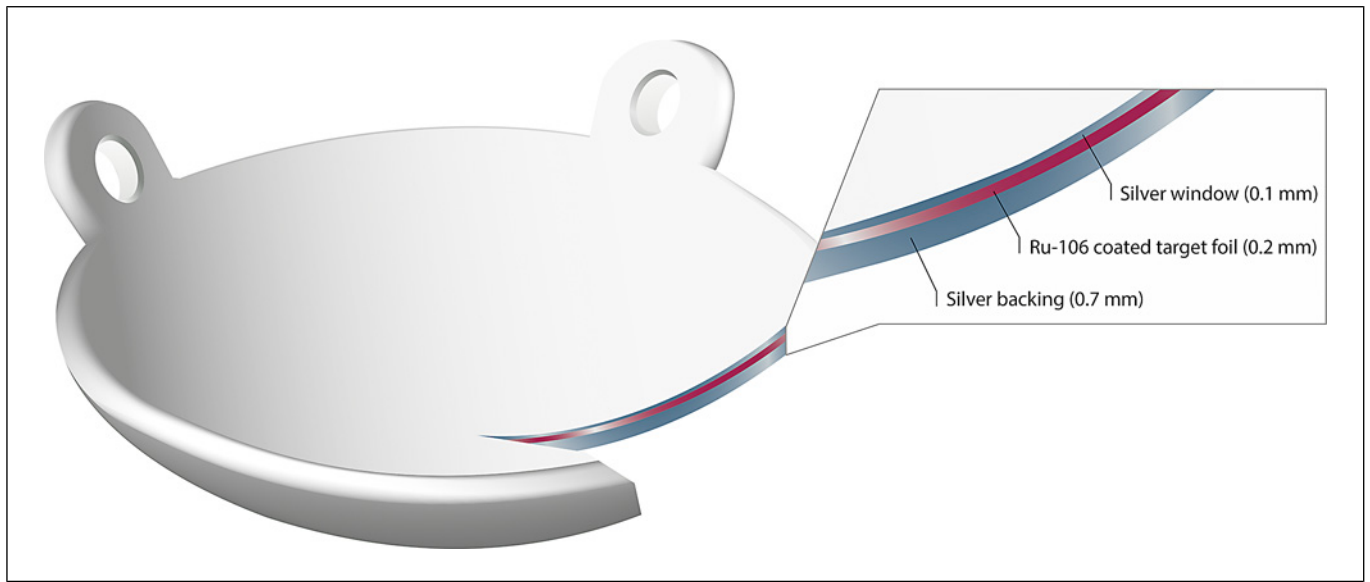


Fig. 2. Silver coating of ruthenium-106 plaques (Courtesy of Eckert & Ziegler Radiopharma GmbH).

carried further onto the eye if care is not taken to dissolve it in excessive amounts of hot water before application. Remnants of silver chloride will be very difficult to remove even with hot water. Drying will not remove excess oxidation agents from LifeClean™, while this will be sufficient to remove excess oxidation agents when PeraSafe™ is used. The rough surface formed when using LifeClean™ will further enhance these differences.

A test was made by using two plates of pure silver (99.9%, K.A. Rasmussen) where each underwent its respective cleaning procedure using either PeraSafe™ or LifeClean™. Through this experiment, we quickly observed a dark layer developing on the silver surface where LifeClean™ was used, while only minor tarnishing of the polished surface was observed where PeraSafe™ was used. Subsequent analysis by X-ray diffraction showed clear formation of silver chloride on the surface with LifeClean™, while no such formation was observed where PeraSafe™ was used. Neither of the surfaces showed any sign of silver oxide remnants.

We performed a spot test on the dried surfaces to check for any remaining oxidative power. The spot test consists of a solution of potassium iodide and starch in water. Any oxidation will lead to formation of iodine and dark color on the cotton swab used. No such formation was observed on the surface where PeraSafe™ was used, and only a very minor stain was observed for the surface where LifeClean™ was used. It is uncertain though if this stain was due to the transfer of dark silver chloride from the surface itself, and hence, the test should be considered inconclusive.

Results

A total number of 6 patients out of 11 treated with ruthenium-106 brachytherapy over a period of 4 months developed a severe inflammatory reaction, and frosted brachytherapy plaques were seen on the removal. During the same period of time, only 1 patient was treated with iodine-125 (at the Department of Ophthalmology, Oslo University Hospital) and 3 patients underwent proton therapy abroad. Although many of the patients with severe inflammatory reactions received antibiotic treatment for a clinically suspected infection initially in addition to routinely given topical combined antibiotics and steroids, the removal of the plaques resulted in clinical improvement. They all underwent treatment according to the radiation dose and time as pre-calculated, although this varied between patients (Table 1). The collaborative investigation of the underlying cause of the inflammatory reaction appearing shortly after plaque insertion revealed a change in the disinfectant agent used for cleaning plaques. The originally used disinfectant agent PeraSafe™ was replaced by LifeClean™. The ruthenium-106 plaques have a silver coating on the external surface as shown in Figure 2. Chlorine-silver reaction with salt deposition on the surface was suspected. The plaques appeared frosted during removal. Initially, we tried soaking the plaques in boiled water and washing off the frosted area of the plaques. However, only the return to the originally used cleaning agent, PeraSafe™, eliminated the occurrence of new cases of the unusual inflammatory reaction. A test was

performed by using two plates of pure silver where each underwent its respective cleaning procedure using either PeraSafe™ or LifeClean™. The results demonstrated formation of silver chloride on the surface with LifeClean™, while no such formation was observed with PeraSafe™.

Discussion

Brachytherapy requires accurate positioning of the plaque to cover the tumor base, and the plaques emit different ionizing radiations dependent on the isotope used. The inflammatory reaction seen in 55% in patients treated with ruthenium-106 for uveal melanoma over a 4-month period was most likely due to chlorine-silver reaction with salt deposition. Ruthenium-106 brachytherapy plaques have been used at Oslo University Hospital for 3 decades; never such unusual, severe local reaction related to brachytherapy plaque treatment has been seen at our hospital. Only 1 patient with uveal melanoma was treated with iodine-125 during this period. The iodine-125 seeds do not have a silver coating but are placed in a silastic insert glued in a gold plaque. Rinsing the ruthenium-106 plaques several times in warm water, thereby washing off the chlorine salt deposits, reduced the salt deposits, but reverting to PeraSafe™ eliminated further cases. Further chemical testing of PeraSafe™ and LifeClean™ on silver surfaces resulted in formation of silver chloride with only LifeClean™. Based on these observations, we recommend being cautious using disinfectants based on chlorine-containing compounds and avoid using strong acidic solutions. The brachytherapy plaque may consist of different metals, and therefore further preclinical tests may be necessary to test how these metals react with cleaning solutions.

A change in disinfectant agent for brachytherapy plaques may give adverse inflammatory reactions. A careful assessment and multidisciplinary collaboration may be needed to aid in diagnosis of this adverse reaction. New steps in the procedures should be reported

to all engaged in the treatment chain. Retrospectively, if everyone in the treatment chain were aware of the new routine, they would have been able to detect abnormal reactions earlier.

Statements of Ethics

Informed consent to publish this case report has been obtained from the patients in writing. This case series was approved by the institutional data protection officer at Oslo University Hospital (23/05463), and there was no need for ethical approval. The CARE Checklist for this case report is available as online supplementary material (for all online suppl. material, see <https://doi.org/10.1159/000548234>).

Conflict of Interest Statement

The following authors have no financial disclosures or any conflicts of interest: J.N., T.B., O.N., B.R., R.F., and N.E.

Funding Sources

The authors declares no funding was obtained or used for this study.

Author Contributions

Conceptualization: T.B. and J.N. Visualization: N.E., J.N., and T.B. Methodology and investigation: O.N. and B.R. Supervision and writing – original draft: J.N. Writing – review and editing: N.E., T.B., O.N., B.R., R.F., and J.N.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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